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TNQ

QUARTERLY EXAMINATION - 2025

12 - STD

BIOLOGY

Time : 3.00 Hrs

Marks : 70

BIO – BOTANY

Marks : 35

I Choose the correct Answers. 8 x 1 = 8

1. Size of pollen grain in Family Cucurbitaceae is
   (a) 10 micrometre (b). 20 micrometer. (c). 200 micrometer. (d). 2000 micrometer.
2. In majority of plants pollen is liberated at.
   (a). 1 celled stage (b). 2 celled stage (c). 3 celled stage (d). 4 celled stage.
3. The dominant epistasis is
   (a). 12:3:1 (b). 9:3:3:1 (c). 9:3:4 (d). 9:6:1
4. The genes controlling the seven pea characters studied by Mendel are known to be located on how many different chromosomes.
   (a). Seven (b). Four (c). Six (d). Five
5. Changing the coden AGC to AGA represents. (a). missense mutation.
   (b). nonsense mutation. (c). frame shift mutation. (d). deletion mutation.
6. EcoRI cleaves DNA at
   (a). AGGGTT (b). GTATATC (c). GAATTC (d). TATAGC
7. Virus free plants are developed from
   (a). Organ culture (b). Meristem culture
   (c). Protoplast culture (d). Cell suspension culture.
8. Solidifying agent used in plant tissue culture is
   (a). Nicotinic acid (b). Cobaltous chloride (c) EDTA (d).Agar

II Answer any four questions. 4 x 2 = 8

9. Write short note on Pollen kitt.
10. Define Lethal genes. Give an example.
11. What is Colchicine?
12. What is bioremediation? Give an example.
13. What do you mean Embryoids?
14. What is Vestigial?

III Answer any three questions. Question No. 19 is Compulsory. 3 x 3 = 9

15. Draw the structure of an ovule and label the parts.
16. What is Atavism? Give an example.
17. What is gene mapping?
18. Write short notes on PCR.
19. What is Cryopreservation?

IV Answer all the questions. 2 x 5 = 10

20. (a). What is endosperm. Explain the types. **(OR)**
)b). Describe Dihybrid Cross.
21. (a). Define Crossing over. Explain the mechanism of crossing over. **(OR)**
    (b). Mention the application of Biotechnology.

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BIO - ZOOLOGY (Marks : 35)

PART - I

I **Note : i) Answer all the questions. ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer** 8 x 1 = 8

1. The purpose of biological treatment of waste water is to ..... .
   a) Reduce sedimentation b) Increase sedimentation c) Reduce BOD d) Increase BOD
2. Allergy involves
   a) IgG b) IgE c) IgA d) IgM
3. The wings of birds and butterflies is an example of
   a) adaptive radiation b) convergent evolution c) divergent evolution d) variation
4. Which of the following is the correct sequence of event with reference to the central dogma?
   a) Transcription, Translation, Replication b) Transcription, Replication, Translation
   c) Duplication, Translation, Transcription d) Replication, Transcription, Translation
5. Trisomic condition of chromosome - 21 results in
   a) Patau's syndrome b) Down syndrome
   c) Klinefelter's syndrome d) Turner's syndrome
6. ..... is the surgical sterilization in women.
   a) Vasectomy b) Diaphragms c) Tubectomy d) Cervical caps
7. The fusion of morphological and physiological identical gametes (isogametes) is called .....
   a) Merogamy b) Isogamy c) Anisogamy d) Hologamy
8. **Find the wrongly matched pair.**
   a) Bleeding phase - fall in oestrogen and progesterone
   b) Follicular phase - rise in oestrogen
   c) Luteal phase - rise in FSH level
   d) Ovulatory phase - LH surge

PART - II

II **Note : Answer any four of the following questions.** 4 X 2 = 8

9. Name an organism where cell division is itself a mode of reproduction.
10. What is inhibin? State its functions.
11. Expand the following. a) IUT b) ZIFT
12. Differentiate -template strand and coding strand.
13. What are interferons? Mention their role.
14. What is SCP?

PART - III

III **Note : Answer any three of the following questions. Question No. 19 is compulsory.**

15. Define the following terms : a) zymology b) super bug 3 X 3 = 9
16. Explain the three major categories in which fossilization occur.
17. Differentiate focticide and infanticide.
18. Placenta is an endocrine tissue. Justify.
19. What is Barr body?

PART - IV

IV **Note : Answer all the questions.** 2 X 5 = 10

20. a) Explain the various phases of the menstrual cycle. ***(OR)***
    b) What are the applications of karyotyping?
21. a) Explain the classical model of Lac Operon, proposed by Jacob and Monod. ***(OR)***
    b) Explain the structure of HIV with a suitable diagram.

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